
Coarsening Confers Robustness: Causal Bayesian Optimization under DAG Misspecification

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Abstract

Causal Bayesian optimization (CBO) uses a causal graph to build informative priors and prune the space of candidate interventions, but requires the entire DAG to be specified correctly, and robustness to structural misspecification is a recognized open problem of the field. We propose coarsening as an input contract: the practitioner supplies only a partition of the manipulable variables and the cluster-level (quotient) graph, and we prove that any optimizer whose graph-dependent components factor through this quotient is invariant (exploration sets, priors, surrogates, the entire trajectory) to every misspecification confined within a cluster. We instantiate the contract for the three main families of the field: QCBO, arm-based quotient CBO with complete identification on the C-DAG; QDCBO, which quotients dynamic CBO slice by slice; and QMCBO, which coarsens model-based CBO’s mechanism factorization. Each is byte-for-byte invariant under within-cluster edge errors that measurably shift its full-graph counterpart, and coarsening concentrates the intervention budget. The price is resolution: when the optimum splits a cluster, a fixed partition pays linear-in- T regret, which plateau-triggered refinement down a declared hierarchy turns into a bounded transient. The trade is intrinsic: sub-cluster interventions are provably not identifiable from quotient-level inputs alone: resolution must be bought with exposure.

1 Introduction

Causal Bayesian optimization [Aglietti et al., 2020] unifies two ideas: when the variable Y we wish to optimize sits inside a causal system, the right search space is the space of interventions $\text{do}(\mathbf{X}_s = \mathbf{x}_s)$, and observational data plus the causal graph yield informative priors over the interventional response surfaces. The catch is the input requirement: CBO and its descendants [Aglietti et al., 2021, 2023, Sussex et al., 2023] assume the full DAG is specified correctly, and the survey of Anonymous [2026] ranks robustness to causal-assumption violations first among the field’s open problems, with stress tests showing that method rankings can change substantially when only the assumed graph is perturbed. We study *graph misspecification*: the learner is given, and trusts, a single graph that differs from the truth (as opposed to graph uncertainty, where the learner maintains a posterior). Our thesis is that this fragility is best addressed by committing to less structure of a more reliable kind.

A wrong edge is costly. The assumed graph enters a CBO algorithm only through its derived causal-effect estimators, and it corrupts them in two ways: it biases the do-calculus prior, and it can change the search set itself, possibly removing the candidate that contains the optimum (Remark 2). Either way the damage is a functional of the wrong edge: an estimator that cannot see an edge cannot be damaged by it.

Coarser commitments decide less, so less can be wrong. A cluster-level structural belief encodes strictly fewer decisions than a full DAG: one aggregate judgement between two groups of variables stands in for a bundle of edge-level judgements the modeler is never asked to make. Both ends of this spectrum are served: full-DAG CBO at one end, methods that learn structure online at heavy per-trial cost at the other [Mukherjee et al., 2024, Branchini et al., 2023]. The practical middle, reliable cluster-level structure, has a developed identification theory [Anand et al., 2023,

Ferreira & Assaad, 2025, Wahl et al., 2024] but, to our knowledge, no optimization method.

Quotient CBO. QCBO fills this gap. The practitioner supplies a partition Π of the manipulable variables and the C-DAG $\mathcal{G}^\Pi$ over its clusters (when a fine-grained hypothesis graph is available, $\mathcal{G}^\Pi$ is its quotient, with bidirected edges absorbing latent confounding; Lee et al., 2019). QCBO explores cluster-level interventions whose effects are identifiable from $\mathcal{G}^\Pi$, with identified cluster-level do-calculus estimates as GP prior means. This input never looks inside a cluster: an edge error confined to a cluster cannot even be represented, let alone affect the optimizer.

Contributions.

- **Robustness as an input contract.** Quotient invariance is *method-agnostic* (Theorem 1): any optimizer whose graph-dependent components factor through $(\Pi, \mathcal{G}^\Pi)$ is invariant to arbitrary within-cluster edge errors. We instantiate the contract for the arm-based, dynamic, and model-based families (QCBO, QDCBO, QMCBO; §4).
- **Sample efficiency.** Coarsening concentrates a fixed budget on fewer identifiable cluster-level arms; Prop. 3 characterizes its cost.

2 Background

A structural causal model $\mathcal{M} = \langle \mathbf{U}, \mathbf{V}, F, P(\mathbf{U}) \rangle$ induces a distribution over observables $\mathbf{V}$ and, for any $\mathbf{X}_s \subseteq \mathbf{V}$, interventional distributions $P(\cdot \mid \text{do}(\mathbf{X}_s = \mathbf{x}_s))$ obtained by replacing the mechanisms of $\mathbf{X}_s$ with constants [Pearl, 2009]. The induced graph $\mathcal{G}$ has a directed edge per functional argument and a bidirected edge per shared latent; projecting out latents yields an ADMG over $\mathbf{V}$ [Lee et al., 2019].

Causal Bayesian optimization. Given manipulable variables $\mathbf{X} \subseteq \mathbf{V}$ and target Y , CBO [Aglietti et al., 2020] solves

$$(\mathbf{X}_s^*, \mathbf{x}_s^*) = \arg \min_{\substack{\mathbf{X}_s \in \mathbf{ES} \\ \mathbf{x}_s \in D(\mathbf{X}_s)}} \mathbb{E}[Y \mid \text{do}(\mathbf{X}_s = \mathbf{x}_s)], \quad (1)$$

where $D(\mathbf{X}_s)$ is the interventional domain, maintaining one GP per candidate set $\mathbf{X}_s$ in an exploration set $\mathbf{ES}$ (each a continuous BO sub-problem over $\mathbf{x}_s$), with prior mean the do-calculus estimate from observational data and the assumed graph, and selection by expected improvement [Garnett, 2023]. The exploration set is pruned structurally (MIS/POMIS; Lee & Bareinboim, 2018). Both the pruning and the priors are functions of the assumed graph, which is where misspecification enters.

Coarsening and C-DAGs.

Definition 1 (Coarsening [Madaleno et al., 2026a]). *Given a DAG $\mathcal{G}$ on $\mathbf{V}$, a partition $\Pi = \{C_1, \dots, C_K\}$ of $\mathbf{V}$ induces the surjection $\chi : \mathbf{V} \rightarrow \Pi$ mapping each vertex to its cluster, and the quotient graph $\mathcal{G}^\Pi$ with vertex set Π and edge set $\{\chi(V_i) \rightarrow \chi(V_j) \mid V_i \rightarrow V_j \in \mathcal{G}, \chi(V_i) \neq \chi(V_j)\}$. If $\mathcal{G}^\Pi$ is acyclic, it is a **coarsening** of $\mathcal{G}$ and Π is valid.*

For ADMGs the quotient additionally carries $C_k \leftrightarrow C_\ell$ iff some $V_i \in C_k, V_j \in C_\ell$ share a bidirected edge. The result is a C-DAG in the sense of Anand et al. [2023]: do-calculus on it is sound and complete for the class of fine graphs that quotient to it. Markov properties transfer to the cluster level, faithfulness does not [Wahl et al., 2024].

Evaluation metrics. We score trajectories by GAP, introduced with DCBO [Aglietti et al., 2021], and its path-aware variant PA-GAP [Anonymous, 2026]. Let y_0 be the initial incumbent, y_t^{best} the best value found up to trial t , and $\rho_t = \min\{(y_0 - y_t^{\text{best}})/(y_0 - y^*), 1\}$ the clipped improvement ratio toward the oracle optimum y^* (minimization shown). With t^* the first trial attaining the final ratio ρ_T ,

$$\text{GAP} = \frac{\rho_T + (T - t^*)/T}{1 + (T - 1)/T}, \quad \text{PA-GAP} = \frac{1}{T} \sum_{t=1}^T \rho_t \frac{T - t + 1}{T}.$$

GAP combines the final improvement with how early it was attained. PA-GAP averages the improvement over the whole run with linearly decaying weights, so it rewards early, steady progress. Both are higher-is-better, and GAP@ k is GAP computed on the first k trials.

3 What coarsening buys

Coarsening is not a property of one algorithm, so we state it as an input contract. An optimizer accepts the pair $(\Pi, \mathcal{G}^\Pi)$ in place of a fine graph, and every graph-dependent component (search space, identification formulas, priors, surrogate architecture, selection rules) is required to be a function of that pair. Any optimizer meeting the contract inherits the invariance below, whatever its family.

Throughout, Π, Π' are manipulable-only partitions. $\Pi_\perp$ is the finest (all-singleton) partition, the bottom of the partition refinement lattice [Madaleno et al., 2026a], and $\Pi' \preceq \Pi$ (Π' refines Π) iff every cluster of Π' is contained in a cluster of Π . Props. 1–3 relate quotients of a common underlying graph. The instantiations of §4 consume only $\mathcal{G}^\Pi$ (Assumption 1).

Theorem 1 (Quotient invariance is method-agnostic). *Let $\mathcal{A}$ be any causal Bayesian optimizer whose*

graph-dependent components (search space, identification formulas, priors, surrogate architecture, and selection rules) depend on the structural hypothesis $\mathcal{G}$ only through the pair $(\Pi, \mathcal{G}^\Pi)$, where $\mathcal{G}^\Pi$ is the quotient of $\mathcal{G}$'s latent projection under a valid Π . Then for any two acyclic hypotheses $\mathcal{G}, \mathcal{G}'$ with $(\mathcal{G}')^\Pi = \mathcal{G}^\Pi$, and the same data, seeds, and hyperparameters, $\mathcal{A}$ produces identical trajectories under $\mathcal{G}$ and $\mathcal{G}'$. In particular, this holds for every edit (insertion, deletion, or reversal of directed or bidirected edges) confined within a single cluster, and for inter-cluster edits that are redundant in the quotient. (Proof: App. A)

Remark 1 (Exact scope). A quotient-level input cannot represent an intra-cluster hypothesis, so no intra-cluster mistake can affect the optimizer. The precise condition is that the perturbed edge be invisible in the quotient: intra-cluster edges, and inter-cluster edges made redundant by a parallel edge. A quotient-visible edge is not protected (§5.3).

Remark 2 (How a wrong graph enters the algorithm). A wrong assumed graph reaches any causal Bayesian optimizer through its derived causal-effect estimators, in two ways. Prior bias: the do-calculus prior mean of any arm whose estimand references the edge is biased even though the true effect is unchanged, a convergence cost paid in GAP and PA-GAP. Search-set change: the exploration set is itself derived from the graph, by (PO)MIS pruning in CBO [Lee & Bareinboim, 2018, Aglietti et al., 2020] and the identifiability gate (Eq. 3) in QCBO, so a wrong graph can flip a verdict and remove an arm (or admit a spurious one). Coarsening closes both channels whenever the edges are quotient-invisible (Theorem 1): the quotient never references the edge, so no estimator it forms is biased and no verdict it makes is flipped.

Proposition 1 (Validity is not refinement-monotone). There exist a valid Π and a refinement $\Pi' \preceq \Pi$ whose quotient is cyclic; validity must be re-verified after each refinement (it remains decidable from cluster-level structure alone).

Proposition 2 (Monotonicity under valid refinement). Let $\Pi' \preceq \Pi$ with both valid and the cluster budget uncapped ($m=K$). Then (i) $\mathbf{ES}_0^\Pi \subseteq \mathbf{ES}_0^{\Pi'}$; (ii) every $\mathbf{X}_s \in \mathbf{ES}^\Pi$ is also in $\mathbf{ES}^{\Pi'}$, with the same identification functional.

Proposition 3 (Price of coarsening). Let $\mu^*(\mathbf{X}_s) = \min_{\mathbf{x}_s \in D(\mathbf{X}_s)} \mathbb{E}[Y \mid \text{do}(\mathbf{X}_s = \mathbf{x}_s)]$ and $V(\Pi) = \min_{\mathbf{X}_s \in \mathbf{ES}_0^\Pi} \mu^*(\mathbf{X}_s)$, with the cluster budget uncapped ($m=K$). For any valid $\Pi' \preceq \Pi$, $V(\Pi_\perp) \leq V(\Pi')$ $\leq V(\Pi)$, and $V(\Pi) = V(\Pi_\perp)$ iff some optimizer $\mathbf{X}_s^*$ of Eq. (1) is a union of Π -clusters.

Proposition 4 (When coarsening is free). Assume (i) full-support intervention domains and (ii) cluster-contained confounding: in $\mathcal{G}^*$, every backdoor path

from a cluster member to Y passes through another member of the same cluster. Then $V(\Pi) = V(\Pi_\perp)$ for every valid Π : setting the remaining members to the values they would naturally have taken changes nothing (consistency), so any sub-cluster arm's value is a mixture over whole-cluster interventions, whose minimum the cluster arm attains. A union-of-clusters optimizer of Eq. (1) then always exists and the price of Prop. 3 is zero. Both premises are necessary (App. A: a policy-box example and a member-target confounding example, each with strictly positive price).

Corollary 1 (The price is a finite-budget effect). Under the premises of Prop. 4, if the quotient keeps the cluster arm covering an optimal fine arm identifiable and the arm-level optimizer is consistent with every arm sampled infinitely often, QCBO's incumbent converges to y^* . Coarsening then changes where a finite budget concentrates, never the attainable value.

3.1 Refining the partition: Hierarchical QCBO (HQCBO)

When no optimizer of Eq. (1) is a union of Π -clusters, a fixed-partition run pays $V(\Pi) - y^* > 0$ forever: $R_T/T \rightarrow V(\Pi) - y^*$. Prop. 4 delimits when this can happen at all: a positive price requires a constrained intervention domain (the policy box of CLUSTERBENCH10L) or confounding that escapes the cluster. Note the kind of failure this is. With a correct graph, a full-DAG method always retains an optimal arm, and a biased prior is eventually overridden by interventional data. A fixed coarse partition lacks the sub-cluster arm even when its quotient is correct: the gap is a property of the action space, and no amount of data restores it. (Under a wrong graph a full-DAG method can suffer the same uncorrectable arm loss through a flipped verdict, Remark 2; refinement addresses the action-space kind.) Refinement converts this failure back into a correctable one by treating Π as a starting resolution.

The practitioner may declare a *hierarchy* H : per cluster, an optional sub-partition (default: singletons); no new graph is supplied. HQCBO runs Algorithm 1 at Π and, when the incumbent improves by less than δ over k trials ($k=5$, $\delta=10^{-3}$, tuned once and fixed), splits the incumbent arm's clusters via H , refusing any split that fails the validity re-check (Prop. 1). On acceptance the exploration set is rebuilt and unioned with every explored arm, data carry over keyed by arm (Prop. 2), and only new arms receive the standard n_0 initialization points (App. G). We evaluate a single refinement step.

Remark 3 (Phase-scoped invariance). Any misspecification that is quotient-invisible at every visited partition, in particular inside a cluster that is never split,

leaves the entire HQCBO trajectory unchanged, including the trigger time and the refinement decision (per phase this is Theorem 1, and the trigger is a function of the trajectory and H alone). Refining a cluster spends its invariance: edits inside a split cluster are exposed after the trigger.

Remark 4 (The price becomes a transient). *If the exploration set at some visited partition contains an optimal arm and the base optimizer is consistent on a finite arm set, the incumbent converges to y^* and $R_T \leq t_r r_0 + o(T)$, with t_r the trigger time and r_0 the initial regret, versus $\Theta(T)$ for the fixed partition. The trigger never observes y^* , so it also fires on loss-less partitions. There refinement costs only the new arms’ initialization and, empirically, a small refinement price (§5.4).*

Remark 5 (Splitting as coarse-to-fine MIS search). *(PO)MIS pruning discards arms from the graph alone [Lee & Bareinboim, 2018]. HQCBO descends the partition lattice guided by observed value, stopping at the coarsest visited partition whose identifiable cluster unions contain a value-optimal arm: a value-driven search for a partition exposing a minimal optimal intervention set, restricted to the declared hierarchy and the incumbent’s clusters.*

4 Instantiations: QCBO, QDCBO, and QMCBO

The contract is met by replacing each graph-coupled component of a base optimizer with its quotient counterpart. We instantiate it for the three main families, in increasing order of graph coupling: CBO (the graph prunes arms and builds priors), DCBO (it additionally carries information across time slices), and MCBO (it is the surrogate’s architecture).

4.1 Arm-based: quotient CBO (QCBO)

QCBO uses only cluster-level structure: a partition Π of the manipulable variables $\mathbf{X}$ and a C-DAG $\mathcal{G}^\Pi$ over its clusters and the singleton nodes. Y and every non-manipulable observable stay singletons, and mixing manipulable and non-manipulable variables in one cluster is disallowed, since intervening on a cluster (jointly setting all its members, $\text{do}(C_k = \mathbf{c}_k)$) must be executable. When the practitioner instead holds a fine-grained hypothesis graph $\mathcal{G}$, $\mathcal{G}^\Pi$ is its quotient after projecting out hidden confounders [Lee et al., 2019]. QCBO itself never requires the fine graph. The candidate exploration set contains all unions of at most m clusters, where the budget $m \leq K$ caps how many of the K clusters one intervention may touch (uncapped

means $m = K$):

$$\mathbf{ES}_0^\Pi = \left\{ \bigcup_{C_k \in \mathcal{S}} C_k \mid \emptyset \neq \mathcal{S} \subseteq \Pi, |\mathcal{S}| \leq m \right\}. \quad (2)$$

Merging manipulable variables coarsens the action space: QCBO can only set a cluster jointly, never a strict subset of it. This is the mechanism behind both the robustness (§3) and the price (Prop. 3) of coarsening. Algorithm 1 (App. D) summarizes the loop.

Identifiable exploration set and the two-tier prior. For each candidate $\mathbf{X}_s \in \mathbf{ES}_0^\Pi$ we pose the cluster-level query $P(Y \mid \text{do}(\mathbf{X}_s))$ on $\mathcal{G}^\Pi$ and run the complete ID algorithm of Shpitser & Pearl [2006] (via **ananke**; Lee et al., 2023). Following the arm-pruning tradition of structural causal bandits [Lee & Bareinboim, 2018], QCBO *commits to its structural beliefs*: the exploration set keeps exactly the candidates whose effect is identifiable from $\mathcal{G}^\Pi$,

$$\mathbf{ES}^\Pi = \{ \mathbf{X}_s \in \mathbf{ES}_0^\Pi : P(Y \mid \text{do}(\mathbf{X}_s)) \text{ id. from } \mathcal{G}^\Pi \}. \quad (3)$$

This restriction is exact: ID is complete on the C-DAG, and identification from $\mathcal{G}^\Pi$ is sound for every fine graph in its class [Anand et al., 2023].

For $\mathbf{X}_s \in \mathbf{ES}^\Pi$, the GP prior mean is constructed in two steps. The identified formula is instantiated as a pipeline of GP-based plug-ins (backdoor $\rightarrow$ front-door $\rightarrow$ g -computation) on the bidirected-edge expansion of $\mathcal{G}^\Pi$, falling back to the observational mean on the queries the sound-but-incomplete pipeline cannot instantiate:

$$m_s(\mathbf{x}_s) = \begin{cases} \hat{\mathbb{E}}[Y \mid \text{do}(\mathbf{X}_s = \mathbf{x}_s)] & \text{pipeline succeeds,} \\ \bar{Y}_{\text{obs}} & \text{otherwise.} \end{cases} \quad (4)$$

Because clusters are manipulable-only and interventions assign full clusters, the cluster-level formulas expand exactly into fine-grained regressions on observational data.

Assumption 1 (Valid C-DAG; estimator boundary). *The directed part of $\mathcal{G}^\Pi$ is acyclic, which constrains only inter-cluster structure (intra-cluster acyclicity can even be relaxed; Madaleno et al., 2026b), coincides with validity of the coarsening when $\mathcal{G}^\Pi$ is a quotient, and is checkable on the cluster-level object. The ID test (Eq. 3) is complete, while the estimator pipeline behind Eq. (4) is sound but not complete: identifiable queries it misses receive the uninformative tier (in §5 this set is empty).*

Proposition 5 (Identity recovery). *Under $\Pi = \Pi_\perp$, $\mathcal{G}^\Pi = \mathcal{G}^*$ and $\mathbf{ES}_0^\Pi$ is the set of all subsets of $\mathbf{X}$ of size at most m ; QCBO reduces to full-DAG CBO, with the exploration set restricted to the identifiable subsets.*

4.2 Dynamic: quotient DCBO (QDCBO)

Dynamic CBO (DCBO; Aglietti et al., 2021) optimizes a sequence of interventions on a dynamic Bayesian network: each slice carries its own exploration sets and priors, built from within-slice (emission) and cross-slice (transition) mechanism estimates on the assumed temporal graph, and slice t 's optimum feeds slice $t+1$'s priors. The graph enters the per-slice exploration sets, both estimate families, and the propagation structure.

Construction (QDCBO). Quotient the temporal graph slice by slice: one vertex $C_{k,t}$ per cluster per slice, quotient emission edges within a slice, quotient transition edges across slices. Exploration sets are lifted to whole-cluster patterns per slice, and the emission and transition estimates become joint cluster mechanisms, one multi-output GP per cluster arc (as in QMCBO below). Singleton clusters reduce to stock DCBO.

Corollary 2 (QDCBO inherits the contract). *Every graph-dependent component of QDCBO is a function of $(\Pi, \mathcal{G}^\Pi)$ applied per slice, so Theorem 1 applies: quotient-invisible edits, in particular any edit confined to a cluster within a slice, leave its entire trajectory unchanged. At $\Pi = \Pi_\perp$, QDCBO is exactly DCBO, and its reachable intervention set at each slice equals QCBO's cluster-level arms, so the price of coarsening (Prop. 3) applies at each slice.*

4.3 Model-based: quotient MCBO (QMCBO)

Model-based CBO (MCBO; Sussex et al., 2023) couples the surrogate to the graph more tightly than CBO: it fits one GP per *mechanism* and evaluates interventions by composing these GPs along the DAG. The graph is the surrogate's architecture, so a wrong intra-cluster edge corrupts the factorization every prediction flows through.

Construction (QMCBO). Coarsen the factorization: one *joint* mechanism per cluster, $P(\mathbf{X}_C | \mathbf{X}_{\text{pa}(C)})$, modeled as a multi-output GP whose intrinsic coregionalization model (ICM) task kernel couples the cluster's member outputs [Bonilla et al., 2008], composed along the C-DAG with joint within-cluster sampling. Intervention targets are lifted to whole-cluster patterns, and singleton clusters reduce to MCBO's per-node GPs.

Corollary 3 (QMCBO inherits the contract). *Every graph-dependent component of QMCBO is a function of $(\Pi, \mathcal{G}^\Pi)$, so Theorem 1 applies: quotient-invisible edits leave its entire trajectory unchanged. Moreover, (i) at $\Pi = \Pi_\perp$, QMCBO is exactly MCBO; and (ii) QMCBO has no arms, but its reachable intervention*

set (whole-cluster do patterns) equals QCBO's cluster-level arms, so Prop. 3 applies: on a lossy partition, no amount of data recovers y^ .*

Proposition 6 (Soundness of cluster mechanisms). *Assume causal sufficiency and a valid Π . For every union-of-clusters intervention, the cluster-level truncated factorization (the true joint cluster mechanisms composed along the C-DAG with intervened clusters clamped) equals $P(\mathbf{V} | \text{do}(\mathbf{x}_S))$. (Proof: App. A.) The independent-outputs variant (QMCBO-ind), which fits one single-output GP per cluster member given the cluster's parents and thus drops within-cluster output correlation, is its approximation, exact when clusters contain no intra-cluster edges. We ablate the two in §5.*

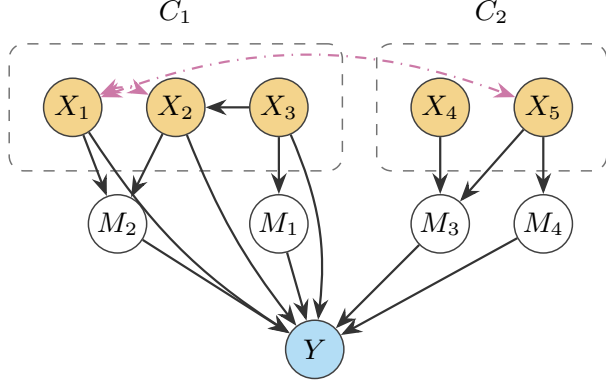
Proposition 7 (Exposure: sub-cluster interventions are not identifiable from cluster mechanisms). *There exist SCMs M_1, M_2 , causally sufficient and valid for the same Π , with identical joint cluster mechanisms and identical interventional distributions for every union-of-clusters intervention, yet $P^{M_1}(Y | \text{do}(\mathbf{x}_A)) \neq P^{M_2}(Y | \text{do}(\mathbf{x}_A))$ for a strict subset $A \subsetneq C$ of some cluster. (Proof: App. A.) Consequently no quotient-respecting method, arm-based, dynamic, or model-based, can evaluate sub-cluster arms from its inputs alone: the exposure trade of §3.1 is a property of the quotient contract itself, not of our refinement procedure.*

5 Experiments

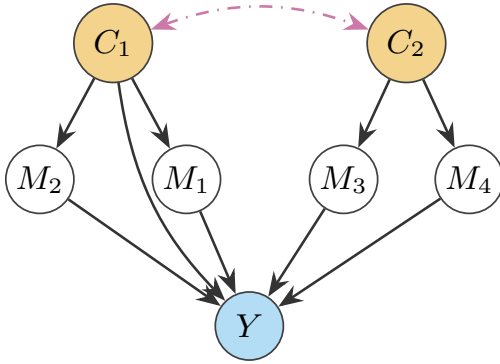
5.1 Setup: head-to-head under misspecification

ClusterBench10. We construct an SCM (Fig. 1) with ten observables: manipulable $X_1, \dots, X_5$, mediators M_1, M_2, M_3, M_4 , target Y , and latent confounders $U_{12} \rightarrow \{X_1, X_2\}$ (intra-cluster) and $U_{15} \rightarrow \{X_1, X_5\}$ (cross-cluster). The coarse partition is $C_1 = \{X_1, X_2, X_3\}$, $C_2 = \{X_4, X_5\}$. Coefficients are chosen so that (i) $\text{do}(X_1)$ is the strongest *singleton* intervention, (ii) the global optimum is $\text{do}(C_1)$ (a union of clusters, so the partition is *lossless* and any robustness gap is not confounded by a price of coarsening), and (iii) $\text{corr}(X_1, X_2) \approx 0.70$ (full specification and oracle checks in App. B).

Fixed-objective protocol. We hold fixed the true SCM, observational data, y^* , seeds, and budget, and inject the misspecification only into the assumed structure each causal method reasons with, verifying that the injection leaves the objective byte-identical under every perturbation (App. C).



(a) fine DAG with coarse partition



(b) quotient C-DAG

Figure 1: **ClusterBench10** (orange: manipulable, blue: target, dash-dotted: latent confounding, dashed boxes: clusters). In the quotient (b), the intra-cluster confounding $X_1 \leftrightarrow X_2$ disappears while the cross-cluster $X_1 \leftrightarrow X_5$ becomes $C_1 \leftrightarrow C_2$. Hence the *intra* bow (adding $X_1 \rightarrow X_2$) is invisible to the quotient, whereas the *inter* bow (adding $X_1 \rightarrow X_5$) completes a cluster-level bow, rendering $\text{do}(C_1)$ non-identifiable and removing it from the coarse exploration set.

Perturbations, methods, and metrics. We perturb the DAG by adding, deleting, or reversing edges, classified as *intra* or *inter* by whether their endpoints share a coarse cluster. The headline perturbation is a *bow*: a directed edge added on a latently-confounded pair, rendering the corresponding do-effect non-identifiable. The arms share one estimator, seeds, and initialization, differing only in the structure each reasons with: BO reads none and optimizes one joint arm under a zero-mean prior, CBO [Aglietti et al., 2020] identifies on the full assumed DAG over (PO)MIS arms (by Prop. 5 it is QCBO at the finest partition, on every graph), and QCBO runs the same algorithm on the quotient over identifiable cluster unions.

We score trajectories with GAP and PA-GAP (§2) and report the paired-by-seed degradation $\Delta = \text{metric}(\text{misspecified}, s) - \text{metric}(\text{correct}, s)$, so per-seed noise cancels, alongside the final value Y_T , the simple regret $r_T = |y_T^{\text{best}} - y^*|$, and the cumulative regret $R_T = \sum_{t=1}^T r_t$ of the incumbent sequence. For GAP a positive Δ means the misspecified run converged faster, and for Y_T under minimization a negative Δ means it reached a better value.

5.2 Coarsening guarantees an unchanged trajectory

We first fix the correct DAG (Table 6): the causal methods exploit the observational prior to converge within a handful of interventions. Non-causal BO, learning the ten-dimensional response from scratch, is far slower and worst in absolute Y .

We then add the intra-cluster bow (add $X_1 \rightarrow X_2$ on the U_{12} -confounded pair) with the objective held fixed. The damage appears in convergence, where Remark 2 predicts it. For QCBO the bow lies inside C_1 : its trajectory, GAP, and PA-GAP are unchanged ($\Delta = 0$ exactly, Theorem 1). CBO deletes every arm the bow touches ($\text{do}(X_1)$ and every superset leaving X_2 unintervened), so its behavior shifts (Table 1).

The same edit with the opposite sign. On this SCM the deletion even *helps*: $\text{do}(X_1)$ is a “decoy”, so removing it concentrates the budget on the larger arms containing the optimum (Fig. 3, App. E). CLUSTERBENCH10D (App. B) moves Y ’s centers so the strongest singleton *is* the global optimum while the coarse partition stays lossless: the identical bow now deletes the cheap optimal arm, and full-DAG CBO’s cumulative regret rises $4.6\times$ (Table 4, App. B), a per-trial cost no data can repair, because the arm itself is gone. QCBO is invariant on both variants. Deleting a decoy accelerates, deleting the optimal arm injures, and which case one is in cannot be known without the very structure in doubt. Theorem 1 removes the question for every quotient-invisible error.

5.3 Which misspecifications are protected?

A single controlled contrast delimits the scope: the *same* operation, adding a directed edge on a latently-confounded pair. The intra bow (add $X_1 \rightarrow X_2$) leaves QCBO unchanged. The inter bow (add $X_1 \rightarrow X_5$, across C_1, C_2) forms a *cluster-level* bow $C_1 \rightarrow C_2$ with $C_1 \leftrightarrow C_2$, making $\text{do}(C_1)$, the arm QCBO relies on, non-identifiable: the guarantee is lost and the trajectory moves off zero (Table 1). The other quotient-visible perturbation, the inter deletion, is unprotected yet unchanged on this SCM (the deleted

edge never enters the identification of an arm the optimizer uses), so quotient-invisibility is *sufficient* for invariance, not necessary. The field also contains the most realistic error of all: omitting the latent confounder $X_1 \leftrightarrow X_2$ (P_7). No verdict flips (the channel is pure prior bias), yet the full-DAG trajectory reshuffles in a seed-dependent direction. The quotient discards intra-cluster bidirected edges, so QCBO is again invariant. The severity dial (Fig. 4, App. E) confirms the pattern across the whole field: QCBO stays at zero across the one-, two-, three-edit intra sweep and leaves zero only under the inter bow.

5.4 When the partition is lossy or wrong

The price of coarsening, realized. CLUSTERBENCH10L shares CLUSTERBENCH10’s structure but strengthens the $X_3 \rightarrow X_2$ mechanism and restricts interventions to a policy box, engineered so the global optimum $\text{do}(X_1, X_3)$ *splits* C_1 (oracle-verified price $V(\Pi) - y^* = 0.95$). The loss is the constrained-domain mechanism of Prop. 4: the mediated path $X_3 \rightarrow X_2$ drives X_2 to values the box forbids the whole-cluster arm to set directly. This is the regime of Prop. 3: CBO reaches y^* while QCBO plateaus a bounded gap above it (realized price 1.03, Table 7, App. E). The price is paid only where coarsening is lossy, and is exactly 0 everywhere else (§5.6).

Refining away the plateau. HQCBO runs the identical coarse configuration with the singleton hierarchy declared for C_1 . The trigger fires at trial 11.1 ± 0.8 (10 seeds), the split passes the validity check, and the warm-started refined phase descends immediately: final $Y = 0.101 \pm 0.001$, indistinguishable from CBO (0.103 ± 0.001), against the QCBO plateau at 1.128 ± 0.016 . Cumulative regret (Remark 4): fixed-partition QCBO accrues $R_{100} = 120.3 \pm 3.1$ and keeps paying forever, while HQCBO’s 32.1 ± 2.6 has stopped growing (Fig. 5, App. G). CBO’s 5.3 ± 0.7 remains the best constant: refinement does not out-run a correctly specified full-DAG optimizer, but it removes fixed-partition QCBO’s linear regret while phase 1 keeps QCBO’s invariance (Remark 3). On the lossless control the trigger still fires and costs nothing (R_{100} : 41.6 ± 5.8 vs 41.8 ± 5.2 ; App. G).

The cost of a wrong partition. Soundness requires the true fine graph to quotient to $\mathcal{G}^\Pi$. App. F stresses that assumption with a fixed, acyclic but *incorrect* C-DAG and finds graceful degradation, not collapse (final $Y = 0.49$ vs 0.13 correct, 0.29 CBO, 1.98 BO): a misspecified *clustering* costs bounded solution quality.

5.5 The contract in the model-based family

We run QMCBO against MCBO through MCBO’s own harness and protocol ($T=100$, 5 seeds) on its two CBO-lineage tasks (TOYGRAPH and PSAGRAP; causal sufficiency holds in both). QMCBO differs from MCBO only in the model’s graph view and the lifted targets.

Invariance (Table 8, App. E). Under an intra-cluster misspecification of the model’s view, QMCBO’s trajectories are byte-identical to their correct-graph runs on 10/10 (env, seed) pairs (Cor. 3 realized), while MCBO’s shift on 8/10, in a direction that is not predictable: on these SCMs the wrong graph sometimes *helped*.

Soundness matters (Table 2). On TOYGRAPH, whose cluster contains the only intra-cluster edge, the joint cluster mechanism reaches the optimum exactly (2.172 ± 0.000) where MCBO attains 1.42 and the independent-outputs ablation 1.26: the within-cluster correlation of Prop. 6 is the missing ingredient. On PSAGRAP, QMCBO edges out MCBO (-4.27 vs -4.35) at a quarter of the wall-clock.

5.6 Each family on its own experiments

We compare each quotient method with its base on the experiments the base method’s own paper ran, in that paper’s own metric. CBO vs QCBO: the 2020 paper’s Toy, Synthetic, and coral-reef ecosystem examples, best attained $E[Y \mid \text{do}]$ over trials at unit cost, 10 seeds. DCBO vs QDCBO: its stationary, independent, and nonstationary DBNs at $T=3$, best Y per time slice, 10 seeds. MCBO vs QMCBO: its ToyGraph and PSAGrap tasks, best attained reward over trials, 5 seeds (the paper’s remaining four problems are function networks, where a partition of function nodes has no causal reading, so we leave them out). Figure 2 overlays the trajectories and Table 3 reports the finals.

The quotient matches or beats its base on six of the eight experiments, including QDCBO on all three of DCBO’s own setups and QMCBO reaching ToyGraph’s optimum exactly where MCBO does not. The two positive prices are exactly where Prop. 3 predicts them: Synthetic, whose optimum $\text{do}(B, D)$ splits the domain partition, and, mildly, Coral. On Toy the coarse arm is not even identifiable from the C-DAG (the cluster forms a bow with Y through the confounder), so QCBO runs it under the uninformative prior and still matches CBO.

Table 1: Scope of the guarantee: paired ΔGAP and ΔY_T (misspecified – correct; mean $\pm$ s.e., 10 seeds; $\Delta Y_T < 0$ means the perturbed run ended closer to y^*). QCB0 is exactly **0** (byte-identical, bold) on every quotient-invisible perturbation and moves only under the inter bow. Full rows in App. E.

Perturbation	Locus	CBO		QCB0	
		ΔGAP	ΔY_T	ΔGAP	ΔY_T
add $X_1 \rightarrow X_2$ (bow)	intra	+0.116 $\pm$ 0.043	−0.190 $\pm$ 0.052	0	0
del $X_3 \rightarrow X_2$	intra	+0.156 $\pm$ 0.039	−0.157 $\pm$ 0.052	0	0
rev $X_3 \rightarrow X_2$	intra	+0.156 $\pm$ 0.039	−0.157 $\pm$ 0.052	0	0
omit $X_1 \leftrightarrow X_2$ (confounder)	intra	+0.098 $\pm$ 0.038	−0.171 $\pm$ 0.039	0	0
sweep $k=1$	intra	+0.116 $\pm$ 0.043	−0.190 $\pm$ 0.052	0	0
sweep $k=2$	intra	+0.120 $\pm$ 0.049	−0.182 $\pm$ 0.055	0	0
sweep $k=3$	intra	+0.120 $\pm$ 0.049	−0.181 $\pm$ 0.055	0	0
del $X_1 \rightarrow M_2$	inter (red.)	+0.122 $\pm$ 0.052	−0.198 $\pm$ 0.048	0	0
del $X_3 \rightarrow M_1$	inter	+0.040 $\pm$ 0.032	−0.091 $\pm$ 0.043	0	0
add $X_1 \rightarrow X_5$ (bow)	inter	+0.007 $\pm$ 0.035	+0.070 $\pm$ 0.073	−0.007 $\pm$ 0.084	+0.538 $\pm$ 0.085

Table 2: QMCBO vs MCBO (its own harness and protocol; 5 seeds, $T=100$). Final Y (all tasks maximized) and wall-clock per unit. QMCBO-ind is the independent-outputs ablation of Prop. 6.

Env	Method	Final Y	s/unit
ToyGraph ($y^*=2.172$)	MCBO	1.422 $\pm$ 0.222	1952
	QMCBO	2.172 $\pm$ 0.000	2563
	QMCBO-ind (ablation)	1.261 $\pm$ 0.160	1181
PSAGraph ($y^*=-4.13$)	MCBO	−4.354 $\pm$ 0.072	8182
	QMCBO	−4.271 $\pm$ 0.126	2753
	QMCBO-ind (ablation)	−4.155 $\pm$ 0.151	2103

Table 3: Realized price of coarsening per family, each pair on the base method’s own experiment suite (final Y , mean $\pm$ s.e.). Price is signed so that positive means the quotient did worse in the task’s direction.

Family	Experiment	Base final	Quotient final	Price	Task
CBO vs QCB0	Toy	−2.09 $\pm$ 0.04	−2.13 $\pm$ 0.03	−0.04	min
CBO vs QCB0	Synthetic	−3.51 $\pm$ 0.03	−1.24 $\pm$ 0.07	+2.26	min
CBO vs QCB0	Coral	36.06 $\pm$ 0.01	36.43 $\pm$ 0.02	+0.37	min
DCBO vs QDCBO	stat ($T=3$)	−6.15 $\pm$ 0.08	−6.40 $\pm$ 0.04	−0.24	min
DCBO vs QDCBO	ind ($T=3$)	−4.98 $\pm$ 0.27	−5.92 $\pm$ 0.07	−0.94	min
DCBO vs QDCBO	nonstat ($T=3$)	7.86 $\pm$ 1.26	7.34 $\pm$ 0.55	−0.52	min
MCBO vs QMCBO	ToyGraph	1.42 $\pm$ 0.22	2.17 $\pm$ 0.00	−0.75	max
MCBO vs QMCBO	PSAGraph	−4.35 $\pm$ 0.07	−4.27 $\pm$ 0.13	−0.08	max

6 Related work

QCB0 occupies an unfilled cell in the CBO landscape: it neither assumes the full DAG nor learns one, but takes a coarser, more reliable input and identifies on the quotient (Table 5, App. E). Methods relaxing the full-graph requirement learn structure online at per-trial cost (GACBO maintains a DAG posterior [Mukherjee et al., 2024], CEO hedges a graph posterior through an entropy-based acquisition [Branchini et al., 2023], Durand et al. [2025] identify $\text{Pa}(Y)$ under a Y -leaf assumption) or assume a known equivalence class [Park et al., 2025]. Averaging only *dilutes* a wrong edge in proportion to posterior mass, whereas

the quotient pays nothing per trial and is exactly invariant to any edit it does not resolve. The identification machinery we rely on is established (C-DAG do-calculus [Anand et al., 2023], C-DMGs over ADMGs [Ferreira & Assaad, 2025], cluster-level Markov properties [Wahl et al., 2024], relaxed admissibility [Yvernes et al., 2025], C-DAGs as discovery background knowledge [Ruiz de Vargas et al., 2025]), but none of it performs optimization. Closest in spirit, Zeitler [2025] coarsen the *outcome* as a cheap proxy [Poloczek et al., 2017, Beckers & Halpern, 2019]. Causal abstraction relates decisions across levels of granularity, and the quotient is such a map, applied to the manipulable *inputs*.

7 Discussion and limitations

Theorem 1 is structural: coarsening acts as a hard regularizer on the hypothesis space, trading resolution (Prop. 3) for immunity to a class of misspecifications. Caveats. (i) The invariance belongs to the *quotient*, and the guarantee is about *behavior*: robustness is read off the CBO-to-QCB0 contrast. (ii) DCBO and MCBO cannot join the fixed-objective seam (per-environment coupling), so QDCBO and QMCBO are each evaluated within their own family’s harness with the base method’s analytic objective held fixed. (iii) Faithfulness does not transfer to the cluster level [Wahl et al., 2024]. (iv) Unit intervention costs only. (v) The identifiability gate costs $\sum_{k \leq m} \binom{K}{k}$ complete-ID checks on the *quotient*, polynomial in clusters for fixed m . (vi) QMCBO assumes causal sufficiency (Prop. 6), as MCBO does. (vii) Full-DAG CBO eventually self-corrects under any wrong graph (Prop. 4 makes this precise), so a fixed partition’s advantages are finite-budget focus and invariance, never limiting value. A

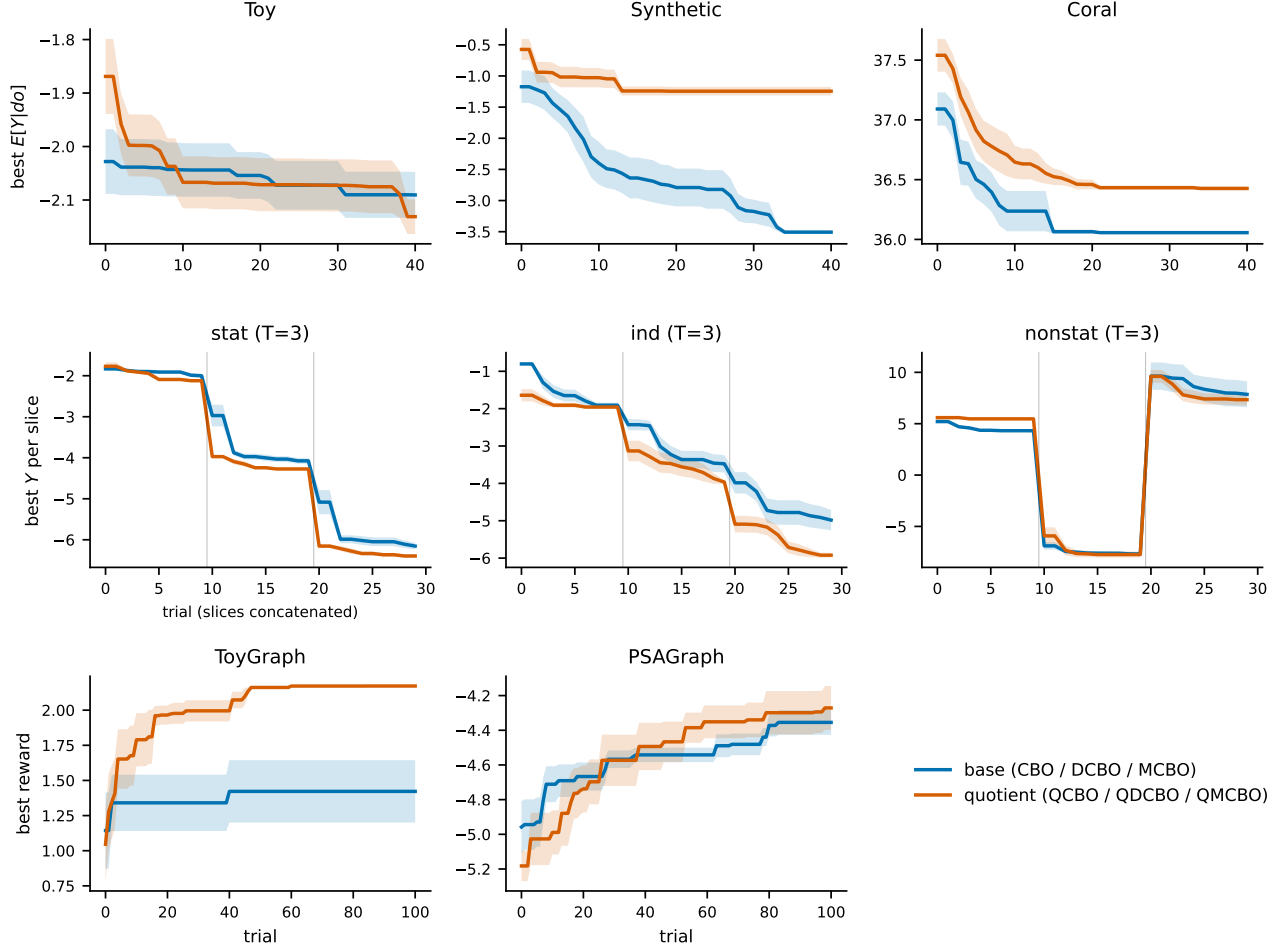


Figure 2: Each quotient method against its base on the base paper’s own experiments and metric (mean $\pm$ s.e.). Top: CBO vs QCBO (best $E[Y | do]$, minimization). Middle: DCBO vs QDCBO (best Y per time slice, trials concatenated across slices). Bottom: MCBO vs QMCBO (best reward, maximization). Dashed line: oracle optimum where known.

rollback rule bounding the lossless-trigger tax, and discovering the hierarchy from the run’s own data, are natural follow-ups.

8 Conclusion

Coarsening runs causal Bayesian optimization on a coarser, more reliable structural commitment with provable invariance to intra-cluster misspecification, at a price that is zero under the conditions of Prop. 4 and bounded-transient under refinement otherwise. The practitioner dials resolution against robustness by choosing how much structure to commit to, in any of the three families.

Acknowledgements

A Proofs

Notation is as in §4.1 and §3.

Proof of Theorem 1. First, $(\mathcal{G}')^\Pi = \mathcal{G}^\Pi$: the latent projection removes only latent vertices, and an edge between two observables is never an interior vertex of a projection path, so $(\mathcal{G}')^*$ and $\mathcal{G}^*$ differ exactly by the modified intra-cluster edges. The quotient keeps a directed (resp. bidirected) cluster-level edge iff some fine edge crosses two distinct clusters, so edits confined within clusters, or redundant across them, leave it unchanged. It remains to show trajectory equality. Write $\mathcal{A}$'s run as a sequence of decisions $d_1, d_2, \dots$ (which arm/action to evaluate, model refits, stopping). By hypothesis each d_t is a deterministic function $d_t = f_t((\Pi, \mathcal{G}^\Pi), \mathcal{D}_{t-1}, \omega_t)$ of the quotient, the data collected so far, and the RNG stream. Under $\mathcal{G}$ and $\mathcal{G}'$ the first arguments coincide. The objective (the true SCM) and the seeds coincide by the fixed-objective protocol, so $\mathcal{D}_0, \omega_1$ coincide, hence d_1 coincides, hence $\mathcal{D}_1$ coincides. Induction over t gives equality of the entire trajectory. $\square$

Proof of Prop. 6. Under causal sufficiency the fine SCM is Markov relative to $\mathcal{G}$ with mutually independent exogenous noises, so for any intervention $\text{do}(\mathbf{x}_S)$ on a union of clusters S the fine truncated factorization gives $P(\mathbf{V} \mid \text{do}(\mathbf{x}_S)) = \prod_{k \notin S} P(X_k \mid \mathbf{X}_{\text{pa}_{\mathcal{G}}(k)})|_{\mathbf{x}_S = \mathbf{x}_S}$. Group the factors by the clusters of Π . For a cluster $C \not\subseteq S$, the product of its members' factors is, by the chain rule applied in any topological order of $\mathcal{G}$ restricted to C , exactly the joint conditional $P(\mathbf{X}_C \mid \mathbf{X}_{\text{pa}(C)})$, where $\text{pa}(C)$, the union of C 's parent clusters, contains every out-of-cluster variable appearing in a member's conditioning set (validity of Π guarantees the quotient is acyclic, so $\text{pa}(C)$ is well defined and precedes C in C-DAG order). The fine factorization therefore equals the product over clusters of the joint cluster mechanisms with the clusters of S clamped at $\mathbf{x}_S$: the cluster-level truncated factorization. Note both uses of sufficiency: independent noises across clusters (no bidirected edges) and the absence of unobserved common causes that would make a member's conditional depend on variables outside $\text{pa}(C) \cup C$. The independent-outputs variant replaces $P(\mathbf{X}_C \mid \mathbf{X}_{\text{pa}(C)})$ by the product of its marginals, exact iff the members are conditionally independent given $\text{pa}(C)$, in particular when C contains no intra-cluster edges. $\square$

Proof of Prop. 7. Take $\mathbf{X} = \{X_2, X_3\}$ as one root cluster C , target $Y = X_2 + \varepsilon_Y$ its child, all noises standard normal and independent (sufficiency holds, and any Π clustering $\{X_2, X_3\}$ is valid). Let M_1 : $X_2 = \varepsilon_2$, $X_3 = aX_2 + \varepsilon_3$; and M_2 : $X_3 = \varepsilon'_3$, $X_2 = a'X_3 + \varepsilon'_2$, with $a' = a/(1+a^2)$, $\text{Var}(\varepsilon'_3) = 1+a^2$, and $\text{Var}(\varepsilon'_2) = 1-a^2/(1+a^2)$. Both induce the same zero-mean Gaussian joint on (X_2, X_3) , with covariance $\begin{pmatrix} 1 & a \\ a & 1+a^2 \end{pmatrix}$, hence the same joint cluster mechanism (here, the same root-cluster distribution), and the same $P(Y \mid \text{do}(\mathbf{x}_C))$ for the whole-cluster intervention (both set X_2 directly). But under the sub-cluster intervention $\text{do}(X_3 = x)$: in M_1 , $X_2 = \varepsilon_2$ is unaffected, so $\mathbb{E}[Y \mid \text{do}(X_3=x)] = 0$ for all x ; in M_2 , $X_2 = a'x$, so $\mathbb{E}[Y \mid \text{do}(X_3=x)] = a'x \neq 0$ for $x \neq 0$, $a \neq 0$. Every quotient-respecting input (the C-DAG, the joint cluster mechanisms, all whole-cluster interventional distributions) coincides between M_1 and M_2 , yet the sub-cluster query differs. No function of those inputs can answer it. $\square$

Proof of Prop. 5. Singleton clusters make the quotient map the identity on $\mathcal{G}^*$, and unions of at most m singletons are exactly the subsets of size at most m . $\square$

Proof of Prop. 1. Take $\mathbf{X} = \{a_1, a_2, b_1, b_2\}$ with fine edges $a_1 \rightarrow b_1$ and $b_2 \rightarrow a_2$. The coarsest partition $\Pi = \{\{a_1, a_2, b_1, b_2\}\}$ is valid: both edges are intra-cluster, so the quotient has a single manipulable node and is acyclic. Its refinement $\Pi' = \{\{a_1, a_2\}, \{b_1, b_2\}\}$ exposes both edges as inter-cluster ($a_1 \rightarrow b_1$ contributes $A \rightarrow B$ and $b_2 \rightarrow a_2$ contributes $B \rightarrow A$), inducing the 2-cycle $A \rightleftarrows B$, so $\mathcal{G}^{\Pi'}$ is cyclic and Π' is invalid. Decidability is immediate from the acyclicity clause of Def. 1. $\square$

Proof of Prop. 2. (i) A union of Π -clusters is a union of Π' -clusters because each Π -cluster is. (ii) Quotienting is compositional: $\mathcal{G}^\Pi$ is the quotient of $\mathcal{G}^{\Pi'}$ under the induced partition of Π' -clusters, so every fine graph compatible with $\mathcal{G}^{\Pi'}$ is compatible with $\mathcal{G}^\Pi$. Identifiability from $\mathcal{G}^\Pi$ means a single functional of the observational distribution equals $\mathbb{E}[Y \mid \text{do}(\mathbf{X}_s)]$ across all SCMs whose graph is compatible with $\mathcal{G}^\Pi$ [Anand et al., 2023]. A fortiori the same functional works across the smaller class compatible with $\mathcal{G}^{\Pi'}$. $\square$

Proof of Prop. 3. By Prop. 2(i), $\mathbf{ES}_0^\Pi \subseteq \mathbf{ES}_0^{\Pi'} \subseteq \mathbf{ES}_0^{\Pi_\perp} = 2^{\mathbf{X}} \setminus \{\emptyset\}$. Minima over larger sets are no larger. Equality holds iff the unrestricted argmin intersects $\mathbf{ES}_0^\Pi$, which is exactly the stated condition. $\square$

Proof of Prop. 4. Fix a cluster C , a strict subset $\mathbf{X}_A \subsetneq C$, a value $\mathbf{a}$, and write $\mathbf{B} = C \setminus \mathbf{X}_A$. Let $\mathbf{B}^{\text{nat}}(\omega)$ be the value $\mathbf{B}$ takes under $\text{do}(\mathbf{X}_A = \mathbf{a})$ at exogenous realization ω . By the consistency of interventions, setting $\mathbf{B}$ to the value it would have taken changes nothing realization by realization, so $Y_\omega(\text{do}(\mathbf{a})) = Y_\omega(\text{do}(\mathbf{a}, \mathbf{B} = \mathbf{B}^{\text{nat}}(\omega)))$. Premise (ii) makes $\mathbf{B}^{\text{nat}}$ independent of the exogenous variables that drive $Y_\omega(\text{do}(\mathbf{a}, \mathbf{b}))$ at fixed $(\mathbf{a}, \mathbf{b})$: every path from a driver of $\mathbf{B}^{\text{nat}}$ into Y other than through C would be a backdoor path from a member to Y escaping the cluster, which (ii) excludes ($\mathbf{X}_A$ is intervened, so paths through it are cut). Taking expectations therefore factorizes into a mixture, $\mathbb{E}[Y \mid \text{do}(\mathbf{a})] = \int \mathbb{E}[Y \mid \text{do}(\mathbf{a}, \mathbf{b})] dP(\mathbf{B}^{\text{nat}} = \mathbf{b} \mid \text{do}(\mathbf{a}))$, and a mixture is bounded below by its best component. Premise (i) makes that component reachable: the whole-cluster arm may set $\mathbf{b}$ anywhere in the natural support, so $\mu^*(C) \leq \mu^*(\mathbf{X}_A)$. Applying the argument per cluster covers arms spanning several clusters, hence $V(\Pi) = V(\Pi_\perp)$, and the minimizing whole-cluster values witness a union-of-clusters optimizer.

Necessity of (i), realized by CLUSTERBENCH10L. The policy box restricts hard interventions to $[-2, 2]$ while the strengthened mechanism $X_2 = 1.5 X_3$ pushes X_2 to ≈ 2.6 , near its bowl center 2.7, when $X_3 \approx 1.7$ is set. The mediated path reaches values the box forbids, so $\text{do}(X_1, X_3)$ strictly beats every in-box whole-cluster intervention: the realized price is 0.95.

Necessity of (ii). Let $C = \{A, B\}$ with latent U , mechanisms $B = U$ and $Y = -BU + (A - 1)^2$, U standard normal. Under $\text{do}(A=1)$ with B free, $\mathbb{E}[Y] = -\mathbb{E}[U^2] = -1$. Under any whole-cluster $\text{do}(A=a, B=b)$, $\mathbb{E}[Y] = -b\mathbb{E}[U] + (a - 1)^2 \geq 0$. Intervening severs the exploitable correlation between B and Y , and the backdoor path $B \leftrightarrow Y$ escapes the cluster, so the price is 1 despite unconstrained domains. $\square$

B ClusterBench10: specification and oracle

Variables and partition. ClusterBench10 has five manipulable variables $\mathbf{X} = \{X_1, \dots, X_5\}$, four non-manipulable mediators $\{M_1, M_2, M_3, M_4\}$, and the target Y , plus two latent confounders $U_{12} \rightarrow \{X_1, X_2\}$ and $U_{15} \rightarrow \{X_1, X_5\}$ (bidirected $X_1 \leftrightarrow X_2$ and $X_1 \leftrightarrow X_5$ in the projected ADMG). The coarse partition is $\Pi = \{C_1, C_2\} \cup \text{singletons}$ with $C_1 = \{X_1, X_2, X_3\}$, $C_2 = \{X_4, X_5\}$. All non-manipulable atoms and Y are singletons. The finest partition $\Pi_\perp$ (all singletons) recovers CBO (Prop. 5).

Structural equations. Roots X_1, X_3, X_4, X_5 carry only exogenous noise (and their latents). The remaining mechanisms are

$$X_2 = 0.3 X_3, \quad M_1 = X_3, \quad M_2 = X_1 + X_2, \quad M_3 = X_4 + X_5, \quad M_4 = X_5,$$

all linear, and the target is a *quadratic bowl* in its parents,

$$Y = \sum_{p \in \text{Pa}(Y)} a_p (p - c_p)^2, \quad \text{Pa}(Y) = \{X_1, X_2, X_3, M_1, M_2, M_3, M_4\},$$

with curvatures $a = (3.0, 1.5, 1.0, 0.5, 0.8, 0.3, 0.3)$ and centres $c = (2, 1, 1, 1, 3, 0, 0)$ over $(X_1, X_2, X_3, M_1, M_2, M_3, M_4)$. The quadratic form places the optimum at an *interior* point (not a box vertex), so unstructured BO cannot solve it in a constant number of evaluations. This is what makes the sample-efficiency comparison non-degenerate. In the observational sampler every node receives additive Gaussian noise ($\sigma = 0.7$, except $\sigma_Y = 1$ and $\sigma_{X_3} = \sigma_{X_4} = \sigma_{X_5} = 1$), and the latents enter their children with unit loadings, $U_{12} \sim \mathcal{N}(0, 1.6^2)$ and $U_{15} \sim \mathcal{N}(0, 1^2)$. The U_{12} scale is what produces $\text{corr}(X_1, X_2) \approx 0.70$. The latent confounders enter only the observational sampler and the declared structure: the noise-free objective every method optimizes is the deterministic SEM above, identical across all methods and perturbations (App. C).

Monte-Carlo oracle (verified at build time). The generator’s oracle minimizes the noise-free objective over each intervention arm’s box and confirms: (C1) $\text{do}(X_1)$ is the strongest *singleton* arm (it lowers Y from the no-intervention value ≈ 23.7 to ≈ 3.6 , a reduction of 20.0, versus 8.1 for $\text{do}(X_2)$ and 5.6 for $\text{do}(X_3)$), so deleting it at the finest partition is a genuine loss; (C2) the global optimum over all subsets is $\text{do}(C_1) = \text{do}(X_1, X_2, X_3)$ at $(X_1, X_2, X_3) = (2, 1, 1)$ with $Y^* = 0$, a *union of clusters*, so the coarse partition is lossless (Prop. 3); C_2 is neutral-to-harmful (its Y -terms are minimized at $X_4 = X_5 = 0$), so a method that intervenes on the full 5-dimensional box wastes two dimensions; (C3) the optimum is interior; and the observational sample has $\text{corr}(X_1, X_2) = 0.70$ via U_{12} , making the bow a genuine confounded misspecification. All perturbed graphs below are acyclic. The build script is `scripts/generate_clusterbench10.py` and the structural premises are asserted in `ccbo/tests/test_clusterbench10_structure.py`.

Table 4: The same intra bow, opposite sign: on CLUSTERBENCH10D the deleted arm *is* the optimum’s cheapest route, and full-DAG CBO’s cumulative regret R_T rises $4.6\times$ (10 seeds, $T=100$, $y^*=0$).

Method	Assumed DAG	Final Y $\downarrow, y^*=0$	R_T $\downarrow$	GAP@100 $\uparrow$
CBO $\equiv$ QCBO-finest	correct	0.001 ± 0.001	2.4 ± 0.6	0.724 ± 0.051
CBO $\equiv$ QCBO-finest	intra bow P_1	0.014 ± 0.005	11.1 ± 2.1	0.724 ± 0.057
QCBO-coarse	either (byte-identical)	0.066 ± 0.023	18.5 ± 4.3	0.809 ± 0.041

Perturbation taxonomy. Each perturbation adds/deletes/reverses one or more edges, classified by whether its endpoints lie within a coarse cluster (*intra*) or across clusters (*inter*), and by whether it is *quotient-invisible* (intra, or inter-but-redundant). The headline contrast is **P1** (add $X_1 \rightarrow X_2$, an intra- C_1 bow on the $X_1 \leftrightarrow X_2$ confounded pair, which makes $\text{do}(X_1)$ non-identifiable at the finest partition) versus **Pic** (add $X_1 \rightarrow X_5$, the same operation across clusters, which forms a cluster-level bow $C_1 \rightarrow C_2$ alongside $C_1 \leftrightarrow C_2$, deleting $\text{do}(C_1)$ at the coarse level). The remaining perturbations are P2 (delete $X_3 \rightarrow X_2$) and P3 (reverse it), both intra- C_1 prior-bias controls; P5 (delete the sole inter edge $X_3 \rightarrow M_1$) and P6 (delete the *redundant* inter edge $X_1 \rightarrow M_2$, with $X_2 \rightarrow M_2$ surviving so the quotient is unchanged); P7 (omit the latent confounder $X_1 \leftrightarrow X_2$ from the assumed graph, the *forgotten confounder*: directed edges and hence all identifiability verdicts are untouched, so this is a pure prior-bias perturbation); and a severity sweep S1–S3 of 1,2,3 simultaneous intra- C_1 edits. QCBO is provably byte-identical on every quotient-invisible perturbation (P1, P2, P3, P6, P7, S1–S3) and is *not* protected on Pic and P5.

ClusterBench10D (the damage companion). Identical structure, mechanisms, noise, and latents: only Y ’s centers move to the values $\text{do}(X_1=2)$ induces in the noise-free SEM, $c = (2, 0, 0, 0, 2, 0, 0)$ over $(X_1, X_2, X_3, M_1, M_2, M_3, M_4)$. Consequently $\text{do}(X_1)$ attains the global optimum $y^*=0$ (the runner-up singleton is ≥ 14), $\text{do}(C_1)$ attains the same value (the coarse partition stays lossless), and $\text{corr}(X_1, X_2) = 0.70$ is unchanged. Build-time oracle checks D1–D3 assert exactly these three premises (`scripts/generate_clusterbench10D.py`). The runner asserts QCBO’s byte-identity under the bow before writing results.

C Fixed-objective protocol: per-method details

The invariant is that every method optimizes the *same* objective, the true SCM’s noise-free `intervention_function`, and only the *structure each method reasons with* is perturbed. We realize this per method as follows.

QCBO (finest and coarse). The objective is read from the original graph’s SEM, while the structure is supplied separately through the assumed graph name (`CoarsenedGraph(..., assumed_graph_name=variant)`). Swapping the assumed graph changes the quotient, the exploration set, the identification formulas, and the prior, but never the SEM that generates outcomes.

CBO (full-DAG identification). CBO is QCBO’s backend at the identity partition (Prop. 5): it runs the same complete identification algorithm, only on the full DAG rather than a quotient. The injection is the same assumed-graph swap as for QCBO (the perturbed structure reaches the exploration set and the identification formulas that build the prior), while outcomes are always drawn from the true SEM. Its trajectory is byte-identical to QCBO’s at the finest partition on every graph.

BO (non-causal control). BO optimizes a single joint intervention on all manipulable variables with a zero-mean GP prior (`Causal_prior=False`): it reads no structure at all, so it is invariant to every perturbation by construction (we run it once and report $\Delta = 0$), at the price of the weakest absolute performance.

Byte-identity verification. For the backend methods the decoupling holds by construction: outcomes are always evaluated on the true graph’s SEM, and the perturbation enters only through the separately-supplied assumed-graph variant. As an independent guard, `seam_ok` evaluates the objective on every singleton arm at random points under the true (P0) versus each perturbed structure and asserts the outcomes match to machine precision. It passes for all perturbations P1–P6 and S1–S3: feeding a method a wrong DAG changes only its causal reasoning, never the problem it is solving.

Algorithm 1 QCBO: Quotient Causal Bayesian Optimization

Require: C-DAG $\mathcal{G}^\Pi$ over the clusters of Π (or a fine hypothesis $\mathcal{G}$ and Π); target Y ; budget T ; observational data $\mathcal{D}^O$; cluster budget m ; optional hierarchy H (§3.1).

- 1: If given $\mathcal{G}$: $\mathcal{G}^* \leftarrow$ latent projection of $\mathcal{G}$; $\mathcal{G}^\Pi \leftarrow$ quotient of $\mathcal{G}^*$ under Π . Check acyclicity of $\mathcal{G}^\Pi$ (Assumption 1).
- 2: Build $\mathbf{ES}^\Pi$ via Eqs. (2),(3) (one complete-ID call per candidate).
- 3: For each $\mathbf{X}_s \in \mathbf{ES}^\Pi$: fit the prior mean m_s via Eq. (4) on $\mathcal{D}^O$.
- 4: Collect initial interventional data $\mathcal{D}_0^I$.
- 5: **for** $t = 1, \dots, T$ **do**
- 6: Update GP posteriors $(m_{s,t}, k_{s,t})$ for all $\mathbf{X}_s \in \mathbf{ES}^\Pi$.
- 7: $(\mathbf{X}_{s_t}, \mathbf{x}_t) \leftarrow \arg \max_{s, \mathbf{x}} \text{El}_t^s(\mathbf{x}) / \text{Co}(\mathbf{X}_s, \mathbf{x})$.
- 8: Intervene, observe y_t ; update $\mathcal{D}_t^I$.
- 9: **if** H declared **and** incumbent plateaued: split the incumbent’s clusters via H , re-check validity, rebuild $\mathbf{ES}^\Pi \cup$ past arms, carry $\mathcal{D}_t^I$ by arm, init new arms (§3.1).
- 10: **end for**
- 11: **return** $(\mathbf{X}_{s_{t^*}}, \mathbf{x}_{t^*})$ where $t^* = \arg \min_t y_t$.

Table 5: Positioning of QCBO against the nearest methods.

Method	What is learned	Structural prior	Search space
CBO [Aglietti et al., 2020]	nothing (oracle DAG)	full DAG	MIS
GACBO [Mukherjee et al., 2024]	posterior over DAGs	DAG family	union of MIS
Durand et al. [Durand et al., 2025]	$\text{Pa}(Y) \subseteq \mathbf{X}$	Y leaf	$\mathbf{X}$
Park et al. [Park et al., 2025]	$\text{PAG} \rightarrow \text{MAG}$	MEC	partial $\mathbf{X}$
SCB [Lee & Bareinboim, 2018]	nothing (oracle DAG)	full DAG	POMIS
QCBO (ours)	nothing	partition + C-DAG	identifiable cluster unions

D Implementation and reproducibility

All methods share one surrogate/acquisition backend (the benchmark’s CBO loop), so comparisons differ only in the assumed structure. **Surrogate.** Each exploration-set arm carries a GPY **GPRegression** model with an RBF kernel (signal variance and lengthscale initialized to 1, ARD off) and observation noise fixed near zero (10^{-10} , matching the deterministic mechanisms). The causal prior supplies the do-calculus estimate as the GP *mean function*, so Δ -metrics isolate the effect of a wrong prior/exploration set from surrogate noise. **Acquisition.** Causal Expected Improvement, maximized over the box-constrained intervention domain by gradient-based L-BFGS. **Protocol.** Unit per-variable intervention costs, 5 initial interventional points per arm, $T=100$ trials, 10 seeds. GAP and PA-GAP are defined in §5.1, and a negative Δ is a degradation throughout. **Invariance tolerance.** Byte-identity of trajectories holds exactly on all seeds of every protected perturbation except one seed of the intra bow, where GP-solver floating-point nondeterminism (independent of the perturbation) perturbs intermediate iterates by $<10^{-8}$ while the recovered optimum is exact. Every quotient-visible perturbation moves trajectories by $\geq 10^{-2}$. We report invariance at this tolerance. **Artifacts.** The ClusterBench10, ClusterBench10L, and ClusterBench10D generators (with their C1–C3 / L1–L3 / D1–D3 oracle checks), the structure-assertion tests, the `seam_ok` guard, every per-method runner, and the table/figure emitters are released, so all reported numbers regenerate from the released SCMs and seeds.

E Full results

Table 9 reports, for every method and every perturbation, the absolute GAP at budgets 20/50/100 and the paired degradation $\Delta\text{GAP}@100$ and $\Delta\text{Final } Y$ (misspecified – correct), mean $\pm$ s.e. over 10 seeds at $T=100$. The CBO row is QCBO at the finest partition $\Pi_\perp$ (Prop. 5). Table 10 reports the corresponding regret: the simple regret r_T of the final incumbent and the cumulative regret R_T of the incumbent sequence (§5.1).

Table 6: Sample efficiency on ClusterBench10 under the *correct* DAG (mean $\pm$ s.e., 10 seeds, $T=100$; higher GAP/PA-GAP and lower r_T/R_T are better).

Method	Final Y $\downarrow, y^*=0$	GAP@20 $\uparrow$ more sample-efficient	GAP@50	GAP@100	r_T $\downarrow$ regret	R_T
BO	1.981 ± 0.26	0.05 ± 0.05	0.15 ± 0.08	0.17 ± 0.09	1.98 ± 0.26	205.6 ± 27.1
CBO	0.290 ± 0.05	0.46 ± 0.06	0.52 ± 0.07	0.56 ± 0.03	0.29 ± 0.05	71.3 ± 5.6
QCBO	0.128 ± 0.02	0.73 ± 0.05	0.89 ± 0.02	0.77 ± 0.07	0.13 ± 0.02	41.8 ± 5.2

Table 7: Realized price of coarsening on CLUSTERBENCH10L (lossy; 10 seeds, $T=100$, $y^*=0.10$). The optimum splits C_1 : QCBO plateaus above y^* by the price of Prop. 3. HQCBO splits C_1 at the trigger and recovers y^* , its R_T a bounded transient.

Method	Final Y $\downarrow$	Final $Y - y^*$ price $\downarrow$	R_T $\downarrow$	GAP@100 $\uparrow$
BO	1.188 ± 0.122	$+1.090$	153.3 ± 32.6	0.765 ± 0.044
CBO	0.103 ± 0.001	$+0.004$	5.3 ± 0.7	0.740 ± 0.045
QCBO-coarse	1.128 ± 0.016	$+1.030$	120.3 ± 3.1	0.802 ± 0.036
QCBO-refine	0.101 ± 0.001	$+0.002$	32.1 ± 2.6	0.694 ± 0.037

F The cost of a wrong partition

QCBO consumes only the C-DAG and never the fine graph, so a wrong *partition* is stressed by quotienting the base structure once under $\{X_1, X_2, X_4\}, \{X_3, X_5\}$ (X_3 and X_4 swapped) to obtain a fixed, acyclic but incorrect C-DAG, and optimizing the true objective through it. The wrong partition alters the exploration set, identification verdicts, and priors, but not the objective. By construction it is invariant to the fine-structure perturbations of §5.3, so it has a single cost, reported in Table 11: final quality degrades ($Y = 0.49$ vs 0.13 correct-partition, 0.29 CBO) while GAP and PA-GAP stay within 0.06 of the correct-partition QCBO run and BO.

G HQCBO across datasets

Table 12 and Fig. 6 give the full picture for HQCBO (§3.1) on the two CLUSTERBENCH variants (10 seeds each; the finest-partition run supplies the CBO row by Prop. 5). Both regimes are visible. *Lossless* (CLUSTERBENCH10): both quotient methods beat CBO on final Y and cumulative regret, and the false trigger (HQCBO cannot observe y^* , so it fires anyway) costs nothing (R_T 41.6 vs QCBO’s 41.8). *Lossy* (CLUSTERBENCH10L): fixed-partition QCBO plateaus with linearly growing regret, while refinement escapes to the CBO-level optimum and bends R_T back. One honest read follows: refinement lowers GAP relative to fixed-partition QCBO (the mid-run split adds arms and re-explores, trading early-budget GAP for final-value and regret recovery), so it is the right default only when the partition is suspected lossy.

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Table 8: Invariance under an intra-cluster misspecification of the model’s graph view (same seeds, deterministic runs): trajectories byte-identical to the correct-graph run, and mean $|\Delta_{\text{final}}|$. QMCBO cannot move (Theorem 1). MCBO does.

Env	Method	byte-identical	$ \Delta_{\text{final}} $
ToyGraph	MCBO	1/5	0.394
	QMCBO	5/5	0.000
PSAGraph	MCBO	1/5	0.261
	QMCBO	5/5	0.000

Table 9: ClusterBench10: full per-method, per-perturbation results (10 seeds, $T=100$). $\text{GAP} \in [0, 1]$, higher is more sample-efficient; Δ columns are paired degradations versus the correct DAG, and **b-id** marks byte-identical trajectories.

Method	Pert.	GAP@20	GAP@50	GAP@100	PA-GAP	ΔGAP	$\Delta\text{PA-GAP}$	ΔY_T	b-id
BO	P0	0.05±0.05	0.15±0.08	0.17±0.09	0.04±0.02	–	–	–	✓
	P1	0.05±0.05	0.15±0.08	0.17±0.09	0.04±0.02	+0.000±0.000	+0.000±0.000	+0.000±0.000	✓
	P2	0.05±0.05	0.15±0.08	0.17±0.09	0.04±0.02	+0.000±0.000	+0.000±0.000	+0.000±0.000	✓
	P3	0.05±0.05	0.15±0.08	0.17±0.09	0.04±0.02	+0.000±0.000	+0.000±0.000	+0.000±0.000	✓
	P7	0.05±0.05	0.15±0.08	0.17±0.09	0.04±0.02	+0.000±0.000	+0.000±0.000	+0.000±0.000	✓
	S1	0.05±0.05	0.15±0.08	0.17±0.09	0.04±0.02	+0.000±0.000	+0.000±0.000	+0.000±0.000	✓
	S2	0.05±0.05	0.15±0.08	0.17±0.09	0.04±0.02	+0.000±0.000	+0.000±0.000	+0.000±0.000	✓
	S3	0.05±0.05	0.15±0.08	0.17±0.09	0.04±0.02	+0.000±0.000	+0.000±0.000	+0.000±0.000	✓
	P6	0.05±0.05	0.15±0.08	0.17±0.09	0.04±0.02	+0.000±0.000	+0.000±0.000	+0.000±0.000	✓
	P5	0.05±0.05	0.15±0.08	0.17±0.09	0.04±0.02	+0.000±0.000	+0.000±0.000	+0.000±0.000	✓
	Pic	0.05±0.05	0.15±0.08	0.17±0.09	0.04±0.02	+0.000±0.000	+0.000±0.000	+0.000±0.000	✓
CBO	P0	0.46±0.06	0.52±0.07	0.56±0.03	0.28±0.03	–	–	–	✓
	P1	0.57±0.05	0.63±0.03	0.68±0.03	0.40±0.01	+0.116±0.043	+0.122±0.024	−0.190±0.052	–
	P2	0.28±0.07	0.58±0.05	0.72±0.04	0.27±0.02	+0.156±0.039	−0.004±0.020	−0.157±0.052	–
	P3	0.28±0.07	0.58±0.05	0.72±0.04	0.27±0.02	+0.156±0.039	−0.004±0.020	−0.157±0.052	–
	P7	0.33±0.06	0.57±0.03	0.66±0.05	0.29±0.02	+0.098±0.038	+0.014±0.022	−0.171±0.039	–
	S1	0.57±0.05	0.63±0.03	0.68±0.03	0.40±0.01	+0.116±0.043	+0.122±0.024	−0.190±0.052	–
	S2	0.53±0.05	0.58±0.04	0.68±0.03	0.36±0.02	+0.120±0.049	+0.079±0.033	−0.182±0.055	–
	S3	0.53±0.05	0.58±0.04	0.68±0.03	0.36±0.02	+0.120±0.049	+0.079±0.033	−0.181±0.055	–
	P6	0.43±0.04	0.62±0.03	0.68±0.04	0.34±0.02	+0.122±0.052	+0.060±0.026	−0.198±0.048	–
	P5	0.46±0.05	0.56±0.05	0.60±0.04	0.30±0.04	+0.040±0.032	+0.024±0.016	−0.091±0.043	–
	Pic	0.47±0.03	0.54±0.04	0.57±0.03	0.35±0.02	+0.007±0.035	+0.071±0.027	+0.070±0.073	–
QCBO	P0	0.73±0.05	0.89±0.02	0.77±0.07	0.48±0.00	–	–	–	✓
	P1	0.73±0.05	0.89±0.02	0.77±0.07	0.48±0.00	+0.000±0.000	+0.000±0.000	+0.000±0.000	✓
	P2	0.73±0.05	0.89±0.02	0.77±0.07	0.48±0.00	+0.000±0.000	+0.000±0.000	+0.000±0.000	✓
	P3	0.73±0.05	0.89±0.02	0.77±0.07	0.48±0.00	+0.000±0.000	+0.000±0.000	+0.000±0.000	✓
	P7	0.73±0.05	0.89±0.02	0.77±0.07	0.48±0.00	+0.000±0.000	+0.000±0.000	+0.000±0.000	✓
	S1	0.73±0.05	0.89±0.02	0.77±0.07	0.48±0.00	+0.000±0.000	+0.000±0.000	+0.000±0.000	✓
	S2	0.73±0.05	0.89±0.02	0.77±0.07	0.48±0.00	+0.000±0.000	+0.000±0.000	+0.000±0.000	✓
	S3	0.73±0.05	0.89±0.02	0.77±0.07	0.48±0.00	+0.000±0.000	+0.000±0.000	+0.000±0.000	✓
	P6	0.73±0.05	0.89±0.02	0.77±0.07	0.48±0.00	+0.000±0.000	+0.000±0.000	+0.000±0.000	✓
	P5	0.73±0.05	0.89±0.02	0.77±0.07	0.48±0.00	+0.000±0.000	+0.000±0.000	+0.000±0.000	✓
	Pic	0.70±0.04	0.73±0.05	0.76±0.05	0.45±0.01	−0.007±0.084	−0.038±0.010	+0.538±0.085	–

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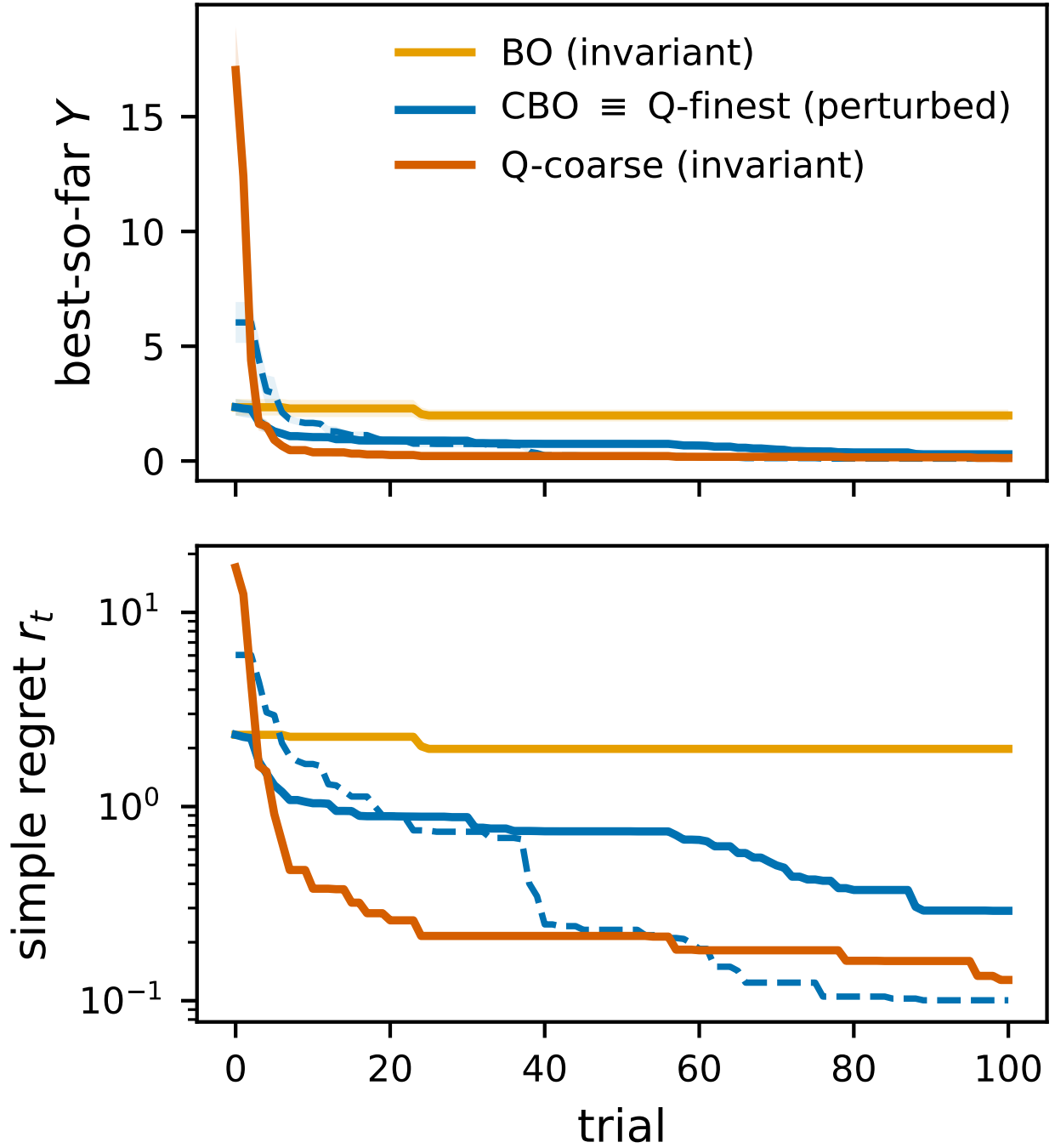


Figure 3: ClusterBench10: correct DAG (solid) vs intra-cluster bow P_1 (dashed). Bands are $\pm$ s.e. over 10 seeds. Top: best-so-far Y . Bottom: simple regret r_t (log scale). QCBO’s and BO’s dashed curves coincide exactly with their solid ones. The full-DAG method’s dashed curve separates from its solid one, but both converge.

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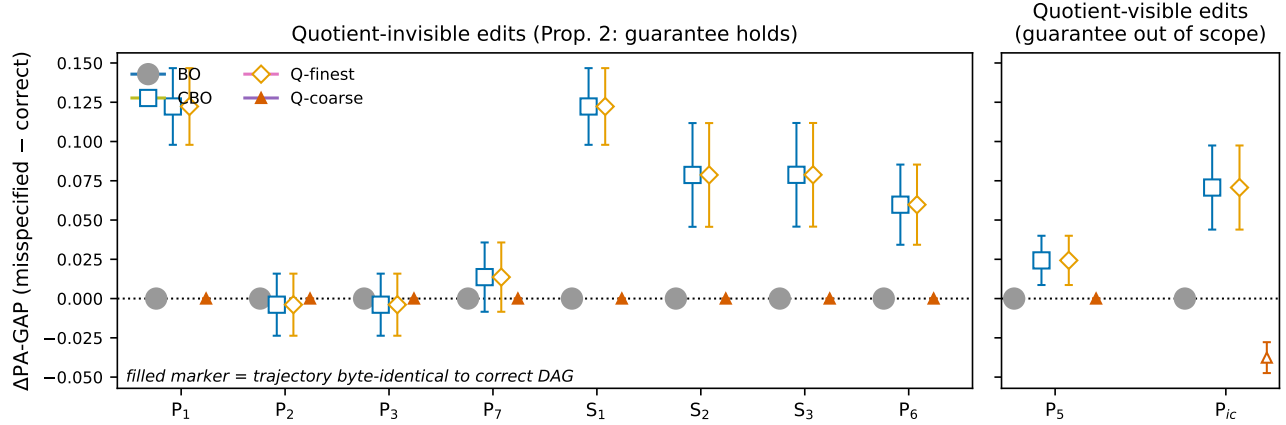


Figure 4: $\Delta\text{PA-GAP}$ (misspecified – correct) for the whole perturbation field, split at the scope boundary of Theorem 1: quotient-*invisible* edits (left) versus quotient-*visible* (right). Filled markers: trajectories byte-identical to the correct-DAG run. QCBO sits at exactly 0 across the left panel and departs only under the inter bow. The full-DAG method moves on every edge it can see.

Table 10: ClusterBench10: simple regret r_T ($\downarrow$) and cumulative incumbent regret R_T ($\downarrow$) per method and perturbation (10 seeds, $T=100$; $y^*=0$).

Pert.	BO		CBO		QCBO	
	r_T	R_T	r_T	R_T	r_T	R_T
P0	1.98 ± 0.26	205.6 ± 27.1	0.29 ± 0.05	71.3 ± 5.6	0.13 ± 0.02	41.8 ± 5.2
P1	1.98 ± 0.26	205.6 ± 27.1	0.10 ± 0.02	65.8 ± 4.6	0.13 ± 0.02	41.8 ± 5.2
P2	1.98 ± 0.26	205.6 ± 27.1	0.13 ± 0.03	62.3 ± 5.4	0.13 ± 0.02	41.8 ± 5.2
P3	1.98 ± 0.26	205.6 ± 27.1	0.13 ± 0.03	62.3 ± 5.4	0.13 ± 0.02	41.8 ± 5.2
P7	1.98 ± 0.26	205.6 ± 27.1	0.12 ± 0.03	58.1 ± 5.5	0.13 ± 0.02	41.8 ± 5.2
S1	1.98 ± 0.26	205.6 ± 27.1	0.10 ± 0.02	65.8 ± 4.6	0.13 ± 0.02	41.8 ± 5.2
S2	1.98 ± 0.26	205.6 ± 27.1	0.11 ± 0.03	98.1 ± 11.6	0.13 ± 0.02	41.8 ± 5.2
S3	1.98 ± 0.26	205.6 ± 27.1	0.11 ± 0.03	98.0 ± 11.6	0.13 ± 0.02	41.8 ± 5.2
P6	1.98 ± 0.26	205.6 ± 27.1	0.09 ± 0.02	43.9 ± 3.9	0.13 ± 0.02	41.8 ± 5.2
P5	1.98 ± 0.26	205.6 ± 27.1	0.20 ± 0.05	59.0 ± 6.4	0.13 ± 0.02	41.8 ± 5.2
Pic	1.98 ± 0.26	205.6 ± 27.1	0.36 ± 0.08	93.2 ± 4.5	0.67 ± 0.07	99.6 ± 4.0

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Table 11: Cost of a *wrong* partition on CLUSTERBENCH10 (correct objective; 10 seeds, $T=100$, $y^*=0$). QCBO-wrongII holds a fixed, acyclic but incorrect C-DAG (base structure under a wrong partition). It loses final quality against the correct partition yet degrades gracefully, staying far above non-causal BO.

Method	Final Y $\downarrow$, $y^*=0$	GAP@100 $\uparrow$ closer to optimum	PA-GAP@100
BO	1.981 ± 0.261	0.175 ± 0.090	0.036 ± 0.020
CBO	0.290 ± 0.054	0.561 ± 0.033	0.278 ± 0.034
QCBO-coarse (correct II)	0.128 ± 0.021	0.769 ± 0.066	0.484 ± 0.003
QCBO-wrongII (wrong II)	0.491 ± 0.088	0.714 ± 0.037	0.464 ± 0.005

Table 12: HQCBO vs BO, CBO, and QCBO across datasets. Final Y (closest to y^* best), GAP@100 ($\uparrow$), cumulative incumbent regret R_T ($\downarrow$); mean $\pm$ s.e., best per group in bold.

Dataset	Method	Final Y $\rightarrow y^*$	GAP@100 $\uparrow$	R_T $\downarrow$
<i>CLUSTERBENCH10</i> (lossless, $y^*=0$)				
	BO	1.981 ± 0.261	0.175 ± 0.090	205.6 ± 27.1
	CBO	0.290 ± 0.054	0.561 ± 0.033	71.3 ± 5.6
	QCBO	0.128 ± 0.021	0.769 ± 0.066	41.8 ± 5.2
	HQCBO	0.070 ± 0.021	0.704 ± 0.059	41.6 ± 5.8
<i>CLUSTERBENCH10L</i> (lossy, $y^*=0.10$)				
	BO	1.188 ± 0.122	0.765 ± 0.044	153.3 ± 32.6
	QCBO	1.128 ± 0.016	0.802 ± 0.036	120.3 ± 3.1
	HQCBO	0.101 ± 0.001	0.694 ± 0.037	32.1 ± 2.6

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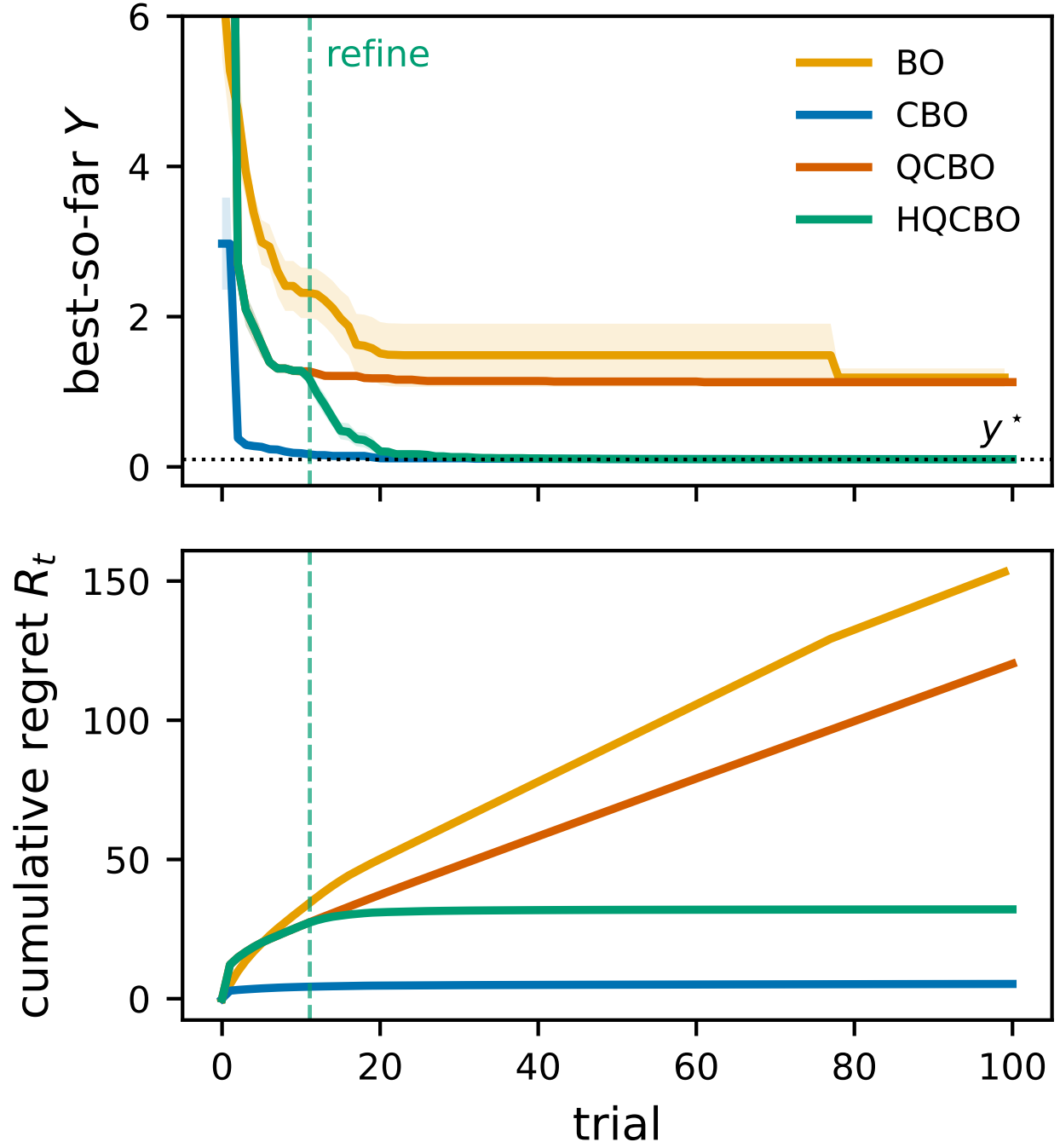


Figure 5: HQCBO on CLUSTERBENCH10L (10 seeds, mean $\pm$ s.e., dashed vertical line = mean trigger). Top: best-so-far Y . HQCBO rides the QCBO trajectory, then splits C_1 and descends to y^* . Bottom: cumulative incumbent regret R_t . Fixed QCBO's is linear in t , HQCBO's flattens after the trigger (Remark 4).

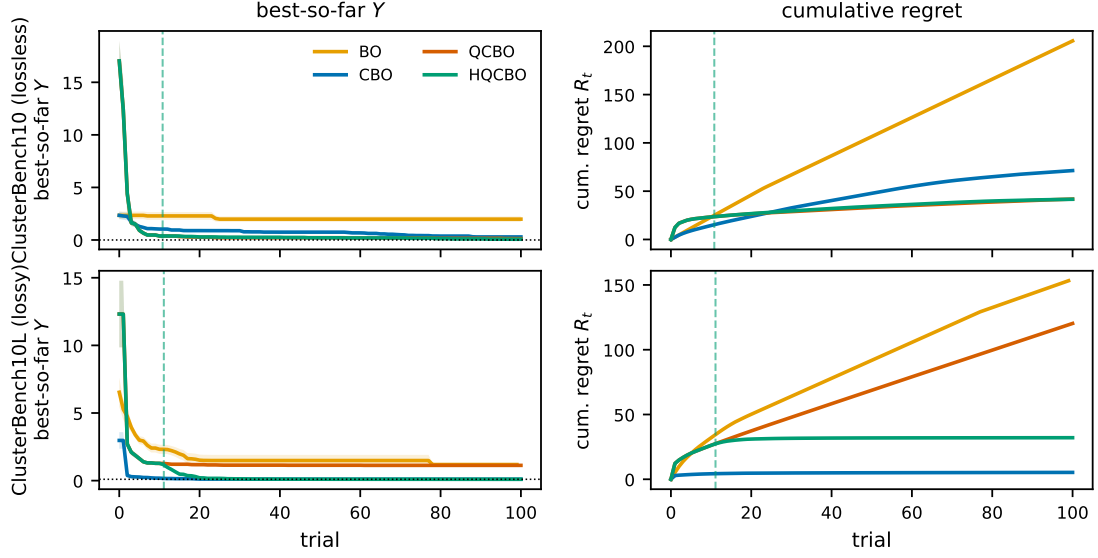


Figure 6: Best-so-far Y (left) and cumulative incumbent regret R_t (right) per dataset (mean $\pm$ s.e., dashed line = mean HQCBO trigger). HQCBO rides QCBO until the trigger, then, where the partition is lossy (CLUSTERBENCH10L), descends and flattens R_t . Where it is lossless the trajectories stay together.